Gauge Group and Generation Structure from the Császár Polyhedron

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Revision note (v6.4): Correction following an audit of October 2026; no new claim is added. (i) Section 5: the alternating-sign operator Y = (−1)^(N+1) × N/3 is withdrawn. It is not affine in the occupation number N, so the three modes carry no definite charge under it, and its labels place the N = 1 and N = 2 sectors, which carry conjugate color representations, in the same one; with either color action one quark sector receives a charge the Standard Model forbids for its color. The charges of the three-mode Fock space are those of Furey's number operator, Q = N/3 (Furey 2015). The audit item left open in v6.2 is closed in the negative: the dressing is not a reparametrization of Furey's choices of ideal and conjugation, and it is not consistent. The novelty of Section 5 is confined to the identification of the three modes with the three Hamiltonian 7-cycles of K_7, verified claim (3) is withdrawn, and the sentence of Section 3 that tied hypercharge to the Fano$^+$ ↔ Fano$^-$ bipartition is corrected to match. (ii) Sections 6.1, 9, 10, 11 and 12: the Cabibbo result was stated as sin θ_C = |V_us| = 3/13. Against PDG 2024 this is excluded at 7.6σ (|V_us|) and 8.5σ (global fit). The ratio agrees with tan θ_C = |V_us/V_ud| to within 0.8σ, but that reading was identified after comparison with data and the construction does not select it, so it is recorded as a post-hoc observation (Conjecture 3), not a verified result. Verified claim (5) is withdrawn from the verified list, and Open Problem (ix) is restated, since Standard Model running of the Cabibbo angle is of order 10^{-4} (Grossman et al. 2022). (iii) Section 3: the 14 faces are the lines of two Fano planes on the seven vertices that share no line, not the lines of one Fano plane and their complements. The automorphism group of the face set is the affine group AGL(1,7), of order 42; PSL(2,7) ≅ GL(3,2), of order 168, is the automorphism group of each Fano plane separately, and only its 21 elements in F_{21} preserve the faces. (iv) Section 8: the chirality statement is given a combinatorial basis; all 42 automorphisms preserve the orientation of the torus, so no realization has a mirror symmetry. (v) Sections 3, 4 and 7.1 and Open Problem (xi): the automorphism group of the quaternions is SO(3), not SU(2). The unit quaternions, Sp(1) ≅ SU(2), act on ℍ by conjugation with kernel {±1}, so SU(2) is the double cover of Aut(ℍ), and it is this unit-quaternion group of each subalgebra that the construction uses. (vi) Section 7.4 stated the spin structure backwards. A torus embedded in $\mathbb{R}^3$ inherits the structure induced from $\mathbb{R}^3$, which is even (Arf invariant 0): antiperiodic around two of the three nonzero classes of $H_1(T^2; \mathbb{Z}_2)$ and periodic around one. The all-periodic structure is the odd one, and no embedding induces it. (vii) Section 2: the uniqueness of the 7-vertex triangulation of the torus is classical (Möbius 1886; see the enumeration in Lutz 2008); Bokowski & Eggert (1991), cited for it before, classify its realizations in $\mathbb{R}^3$. (viii) The title is shortened to what the paper still claims. Four references are added (Grossman et al. 2022; Lutz 2008; Möbius 1886; Particle Data Group 2024), and the reference list now runs 1–27. (ix) Smaller corrections, none of which changes a conclusion. Section 6's subsections are renumbered 6.1–6.3; there had been no Section 6.2 since v5. Sections 6.1, 9 and 12 call 3/13 and its two integers combinatorial rather than topological invariants of the triangulation: the rank of ∂_2 is the number of faces minus one, so it depends on the triangulation. Sections 6.1 and 9 identify the |V_cb| comparison value, 0.0408 ± 0.0014, as PDG 2022's. The abstract and Sections 7.3 and 9 state the centralizer theorem for the Lie algebras, where it is proved: the centralizer of su(3) in g_2 is trivial. The centralizer of the group SU(3) in G_2 is its center $\mathbb{Z}_3$, which is finite, so the conclusion that no weak factor inside G_2 commutes with color stands. Section 7.5 and Open Problem (i): a straight edge has no Frenet–Serret frame, since its curvature vanishes, so each edge carries an adapted orthonormal frame with a chosen normal. Section 7.1: the conformal

structure of a torus is fixed by its geometry, not by its topology. Sections 7 and 12 say, as the abstract does, that the rank-4 problem is addressed rather than resolved, since the tensor-product realization is Open Problem (i). Section 2 cites Heawood (1890) and Ringel & Youngs (1968) for the Heawood bound; the latter had been listed but not cited. Section 7.2 says what W denotes, and a stray "(a)" is removed from Section 5.

Revision note (v6.3): Attribution-completeness micro-revision; no equation, numeric value, claim, conjecture status, or open-problem content is changed. (i) The Section 11 differentiation paragraph now names the Cℓ(8) extension of the sedenion three-generation construction (Gourlay & Gresnigt 2024), completing the mechanism-family coverage of the July 2026 prior-art review, and the corresponding reference is added. (ii) A v6.2 reference-numbering defect is repaired: the final entry (Ringel & Youngs 1968) had retained its pre-v6.2 list number (13) outside the v6.2 renumbering, duplicating an existing label; it is renumbered to its alphabetical position (23). The reference list now runs 1–23. All technical content is unchanged from v6.2.

Revision note (v6.2): Attribution-only revision; no equation, numeric value, claim, conjecture status, or open-problem content is changed. Following a prior-art review against the Furey corpus (July 2026): (i) Section 5 now attributes the Fock-grading/number-operator hypercharge mechanism to Furey (2015; 2016; 2018a) and scopes this paper's contribution to the geometric identification of the three modes with the Hamiltonian 7-cycles of K_7 and, provisionally, the closed-form sign dressing; (ii) the canonical citation Günaydin & Gürsey (1973) is added for SU(3) as the G_2-stabilizer of an imaginary unit (Sections 1, 3, 7.3); (iii) Section 11 gains a differentiation paragraph naming the competing three-generation mechanisms and the principal algebraic alternatives on the rank-4 problem, with the novelty language of Sections 5 and 7 scoped accordingly; (iv) Section 8 gains a note differentiating Conjecture 1 from the algebraic parity-violation results of Furey (2018c) and Furey & Hughes (2022b); (v) Section 7.2 gains one attributed corroboration sentence from the triality identical-action theorem; (vi) nine references are added and the Furey 2018 entries are disambiguated as 2018a/b/c. All technical content is unchanged from v6.1.

Revision note (v6.1): Single-sentence correction to the abstract. The abstract previously stated "These act on independent tensor factors and commute by construction" — language inconsistent with the §7.2 body correction applied in v6. Replaced with a statement that (a) records the centralizer theorem from §7.3 as the proved negative result, and (b) names the tensor-product realization as Open Problem (i), consistent with the rest of the paper. No other content changed from v6.

Revision note (v6): v6 (i) corrects the Section 7.2 tensor-factorization argument — previously stated as automatic from the dimensional independence of the strata — to reflect that the cells of the Császár complex form a chain complex, not independent tensor factors, and that the edges carry a non-trivial color representation ($\Lambda^2(\mathbf{7}) = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{3} \oplus \bar{\mathbf{3}}$ as an su(3)-module); the factorization is now recorded as a requirement to be established. (ii) Adds to Section 7.3 the theorem that the centralizer of color SU(3) in G_2 is trivial ($g_2 = \mathbf{8} \oplus \mathbf{3} \oplus \bar{\mathbf{3}}$, no trivial summand; verified on Der(𝕆)), making rigorous the claim that no commuting weak factor exists inside G_2 and that the additional rank must be sourced externally. (iii) Restates Open Problem (i) as the establishment of a genuine tensor-product realization, shown to reduce to Open Problem (ii) or (xi)(b) according to how the weak SU(2) is realized, and removes the redundant bracket-computation framing; verifiable claim (4) is updated to match. The gauge-group construction, the three-generation and hypercharge derivations, and the Cabibbo-angle derivation are unchanged from v5.

Revision note: v5 adds an explicit quaternion algebra map for edge framings (Section 7.5), a conjecture paragraph on the left-handed coupling gap (Section 7.1), flags the 14 = 12 + 2 dimension count as a structural observation pending an explicit map (Section 7.3), condenses the V_cb material to a note (formerly Section 6.2), softens the Cabibbo physical interpretation to topological language (Section 6.1), and adds Open Problems (x) and (xi). The hypercharge derivation, gauge group construction, and Cabibbo angle derivation are substantively unchanged from v4.

Abstract

We show that the Császár polyhedron — the unique neighborly triangulation of the torus — encodes the Standard Model gauge group SU(3) × SU(2) × U(1) and three fermion generations through its combinatorial topology, carries the three fermionic modes of Furey's number-operator construction of charge, and contains a ratio, 3/13, that matches one reading of the leading CKM mixing parameter. The construction proceeds in four stages: (i) $G_2 = \mathrm{Aut}(\mathbb{O})$ acting on the K_7 vertex structure yields SU(3) via vertex stabilization; (ii) three generations arise from the three Fano lines through any vertex; (iii) the three modes of Furey's number-operator construction, whose charge spectrum Q = N/3 is Furey's (2015), are identified with the three Hamiltonian cycles of K_7; (iv) the ratio of the Fano valence to the rank of the boundary operator in the Császár triangulation is 3/13, which agrees with tan θ_C = |V_us/V_ud| but not with sin θ_C = |V_us|; because the construction does not select the observable, the agreement is recorded as a post-hoc observation.

The rank-4 problem — how the rank-4 gauge group SU(3) × SU(2) × U(1) arises from the rank-2 group G_2 — is addressed by showing that the three gauge actions operate on topological data of different dimension: SU(3) on vertex labels (0-cells), SU(2) on edge spin-framings (1-cells), and U(1) on the surface complex structure (2-cells). That no commuting weak factor exists inside G_2 itself is a theorem: the centralizer of color su(3) in g_2 is trivial ($g_2 = \mathbf{8} \oplus \mathbf{3} \oplus \bar{\mathbf{3}}$ as an su(3)-module, containing no trivial summand), so the additional rank must be sourced externally. Whether the three geometric actions can be jointly realized on a common representation space as a genuine tensor product — which would ensure their commutativity identically — is a substantive open requirement recorded as Open Problem (i) in Section 10.

We further propose that the Császár polyhedron provides a geometric realization of the Dixon algebra $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, with the 0-dimensional vertices carrying $\mathbb{O}$, the 1-dimensional edge framings carrying $\mathbb{H}$, and the 2-dimensional surface complex structure carrying $\mathbb{C}$. A conjecture is offered relating the intrinsic chirality of the Császár embedding to parity violation in the weak sector. No free parameters are introduced. Open problems are explicitly identified.

1. Introduction

The Standard Model gauge group SU(3) × SU(2) × U(1) is among the most precisely tested structures in physics, yet its origin remains unexplained. Several authors have noted connections between division algebras and particle physics. The octonion–quark connection originates with Günaydin & Gürsey (1973), who identified color SU(3) as the stabilizer of an imaginary unit inside G_2. Furey (2015; 2016; 2018a) showed that the algebra $C\ell(6)$, constructed from the octonions, reproduces a single generation of Standard Model fermions with correct hypercharges, with electric charge quantized because it is built from a number operator (Furey 2015). Dixon (1994) introduced the tensor product algebra $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ as the algebraic foundation for the Standard Model. Gürsey & Tze (1996) explored related structures extensively.

This note proposes a specific geometric realization: the Császár polyhedron, whose vertex structure carries the octonion algebra naturally. The argument proceeds in stages: a uniqueness argument selects the polyhedron; vertex stabilization under G_2 yields SU(3); the Fano plane incidence geometry produces three generations; the three Hamiltonian cycles carry the modes of Furey's number-operator charge; and the homological cycle structure of the triangulation produces the ratio 3/13, which Section 6.1 compares with the Cabibbo angle.

2. The Uniqueness Argument

Among compact orientable surfaces, the torus is uniquely selected by the requirement $\chi = 0$ (vanishing Euler characteristic). The Császár polyhedron (1949) is the geometric realization of the complete graph K_7 embedded on the torus. Its uniqueness is classical: the 7-vertex triangulation of the torus, Möbius' torus (Möbius 1886), is the only one with 7 vertices up to isomorphism (see the enumeration in Lutz 2008), and since a neighborly triangulation of the torus has exactly 7 vertices, it is the only neighborly one. Bokowski & Eggert (1991) classify its realizations in $\mathbb{R}^3$. The combinatorial invariants are completely determined: V = 7 (the Heawood bound; Heawood 1890, Ringel & Youngs 1968), E = 21 = C(7,2) (neighborly), F = 14 (Euler: 7 − 21 + 14 = 0), genus g = 1, and $\chi = 0$. Every integer is forced. All 14 faces are triangles (2E = 3F = 42).

3. Gauge Group from Vertex Stabilization

The 7 vertices of K_7 are identified with the 7 imaginary octonion basis elements $\{e_1, \ldots, e_7\}$. With the vertices labelled by $i \in \mathbb{Z}_7$ ($e_7 \equiv e_0$, as in Section 6), the 14 triangular faces form two families of seven, {i, i+1, i+3} and {i, i+2, i+3}. Each family is the line set of a Fano plane PG(2, $\mathbb{F}_2$) on the seven vertices: the first (Fano$^+$) is the set of 7 octonion multiplication triples (e_i e_{i+1} = e_{i+3}), the second (Fano$^-$) shares no line with it, and together they contain every edge exactly twice. (Earlier versions described the second family as the complements of the Fano lines; the complement of a line has four points, not three.) The automorphism group of the 14-face set is the affine group AGL(1,7) = {$x \mapsto ax + b$}, of order 42; its index-2 subgroup F_{21} ($a \in \{1, 2, 4\}$) preserves each Fano plane, and the other coset exchanges them. The group PSL(2,7) ≅ GL(3,2), of order 168, is the automorphism group of each Fano plane separately, not of the face set: only its 21 elements in F_{21} preserve the faces.

The automorphism group of the octonions is the exceptional Lie group G_2, with $\dim(G_2) = 14 = F$. Fix a vertex v_0 (equivalently, fix one imaginary octonion unit). The stabilizer of v_0 in G_2 is SU(3), acting on the 6 remaining vertices which decompose as $3 \oplus \bar{3}$ (outgoing/incoming via circuit directions). This identification — SU(3) as the G_2-stabilizer of a fixed imaginary unit, with the $3 \oplus \bar{3}$ branching on the remaining units — is classical (Günaydin & Gürsey 1973). The coset space $G_2/SU(3) = S^6$ has dimension 6 = V − 1 = vertex degree of K_7. Three Fano$^+$ lines through v_0 define three quaternionic subalgebras. The automorphism group of each is $\mathrm{Aut}(\mathbb{H}) \cong SO(3)$, because the unit quaternions, $Sp(1) \cong SU(2)$, act on $\mathbb{H}$ by conjugation with kernel $\{\pm 1\}$; the unit-quaternion group $Sp(1) \cong SU(2)$ of each subalgebra, the double cover of its automorphism group, provides the weak gauge group. (Versions through v6.3 wrote $\mathrm{Aut}(\mathbb{H}) = SU(2)$.) The Z_2 bipartition Fano$^+ \leftrightarrow$ Fano$^-$ exchanges the two face families. (Versions through v6.3 also said it provides U(1) hypercharge, through the sign dressing of Section 5; that dressing is withdrawn in v6.4, and the U(1) factor is the surface phase of Section 7.) Dimension equation: 14 = 12 + 2, i.e., $\dim(G_2) = \dim[SU(3)\times SU(2)\times U(1)] + \dim H_1(T^2)$.

4. Three Generations from Fano Lines

Three of the 7 Fano lines pass through any given vertex. Each such line defines a quaternionic subalgebra of the octonions. The three lines through v_0 yield three geometrically distinct quaternionic embeddings $\mathbb{H}_1$, $\mathbb{H}_2$, $\mathbb{H}_3 \subset \mathbb{O}$, each carrying its own unit-quaternion group $Sp(1) \cong SU(2)$, the double cover of $\mathrm{Aut}(\mathbb{H}) \cong SO(3)$. This gives exactly three generations of fermions.

The number three is the valence of the Fano plane at any point and cannot be adjusted.

5. Hypercharge from Fock Space Grading

The mechanism of this section — a $2^3 = 8$-dimensional Fock space built from three octonionic fermionic modes, with charge proportional to the occupation number divided by 3, so that charge quantization follows from the integrality of a number operator — is due to Furey (2015), who established it in Clifford-algebraic form with the neutrino playing the role of the vacuum state; the full SU(3) × SU(2) × U(1)(× U(1)) ladder-operator construction on $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ is Furey (2016; 2018a). What this section contributes is a geometric realization of that mechanism on the Császár polyhedron, as follows.

The 21 edges of K_7 decompose into three disjoint Hamiltonian 7-cycles (one for each quadratic residue class mod 7). At the fixed vertex v_0, these three circuits define three fermionic modes with occupation numbers $n\alpha$, $n\beta$, $n\gamma \in \{0,1\}$. The total occupation $N = n\alpha + n\beta + n\gamma$ ranges from 0 to 3, generating a $2^3 = 8$ dimensional Fock space identified with the octonion basis.

The charge operator is Furey's number operator divided by three (Furey 2015):

$$Q = N/3$$

where N is the total circuit occupation number. In Furey's minimal left ideal its eigenvalues are $N = 0 \rightarrow Q = 0$ (neutrino); $N = 1 \rightarrow Q = +1/3$ (anti-down quark, 3 colors); $N = 2 \rightarrow Q = +2/3$ (up quark, 3 colors); $N = 3 \rightarrow Q = +1$ (positron); the complex-conjugate ideal carries the antiparticles, with charge $-N/3$. For SU(2)-singlet states the charge equals the hypercharge. These assignments are Furey's; they are carried here by the cycle modes, not derived. Two structural facts fix their form. First, color acts irreducibly on the three modes, so a U(1) charge commuting with color gives every mode the same charge, and the charge of an N-mode state is then affine in N. Second, the N = 2 states are antisymmetric pairs of N = 1 modes, so the N = 1 and N = 2 sectors carry conjugate color representations, $\bar{3}$ and 3 or 3 and $\bar{3}$ — the $3 \oplus \bar{3}$ of Section 3.

Correction (v6.4). Versions through v6.3 used the operator $Y = (-1)^{(N+1)} \times N/3$, assigned its spectrum $\{0, +1/3, -2/3, +1\}$ to the right-handed neutrino, three down antiquarks, three up antiquarks and the positron, attributed the alternating sign to the complex structure $J = L_{e_0}$, and described the spectrum as the one anticipated by Furey's construction. That operator is withdrawn. It is not affine in N, so the modes carry no definite charge under it; and its labels place the N = 1 and N = 2 sectors in the same color representation ($\bar{3}$), which the conjugate pairing excludes. With either color action on the modes, one quark sector receives a charge that the Standard Model forbids for its color: $(3, +1/3)$ or $(3, -2/3)$. The open audit item recorded in v6.2 — whether the dressing is a reparametrization of Furey's choices of ideal and conjugation — is closed in the negative: each of those choices gives a charge affine in N, and none reproduces the dressing.

The novel content of this section is therefore confined to the identification of the three fermionic modes with the three disjoint Hamiltonian 7-cycles of K_7 (the quadratic-residue classes mod 7) — a geometric statement with no counterpart in the algebraic construction.

6. CKM Quark Mixing from Fano Generation Mismatch

The left-handed (Fano^+) and right-handed (Fano^-) sectors assign different generation labels to the six neighbor vertices of v_0. The Fano^+ doublets are $\{e_1, e_3\}$, $\{e_4, e_5\}$, $\{e_2, e_6\}$; the Fano^- doublets are $\{e_2, e_3\}$, $\{e_4, e_6\}$, $\{e_1, e_5\}$. Every left-handed generation shares exactly one vertex with two different right-handed generations and zero with the third. This structural mismatch is the geometric origin of quark mixing.

6.1 The Cabibbo Angle

The construction's candidate for the leading mixing parameter is the ratio of the Fano point-valence to the rank of the boundary operator in the Császár triangulation:

$$3/13 = 0.23077$$

where 3 is the Fano valence (the number of lines through any vertex, equivalently the number of inter-generation transition channels) and 13 is the rank of the boundary operator ∂_2 in the Császár triangulation.

Homological derivation of 13. The Császár triangulation has F = 14 faces, E = 21 edges, V = 7 vertices. The boundary operator ∂_2: $C_2 \to C_1$ sends each triangular face to the sum of its three boundary edges. The kernel of ∂_2 is $\ker(\partial_2) = H_2(T^2) \cong \mathbb{Z}$, reflecting the single fundamental class of the closed torus. Therefore:

$$rank(\partial_2) = dim(C_2) - dim(ker\ \partial_2) = 14 - 1 = 13$$

This counts the independent contractible 1-cycles on the Császár torus — cycles that bound a region of the surface. The total 1-cycle space has dimension E − V + 1 = 15 (the circuit rank of K_7). Of these, 13 are contractible (boundaries of face-chains) and the remaining 2 are the generators of $H_1(T^2) \cong \mathbb{Z}^2$ — the two non-contractible loops around the torus.

Physical interpretation. The ratio 3/13 is a combinatorial invariant of the Császár triangulation: the numerator is the Fano point-valence (three lines through every vertex, a combinatorial constant of $PG(2,\mathbb{F}_2)$) and the denominator is the rank of the boundary operator ∂_2 (fixed by the face count and the homology of the torus). Both integers are forced by the triangulation; neither is fitted to the data. Whether this ratio is related to Cabibbo mixing at all, and if so through which observable, is the physical question the construction raises but does not yet answer from first principles.

Comparison with data (corrected in v6.4). Earlier versions compared 3/13 with sin θ_C = |V_us| and reported a 2.6% overshoot against 0.2250 ± 0.0007 (PDG 2022). Against PDG 2024 that reading is excluded: 3/13 lies 7.6σ above |V_us| = 0.22431 ± 0.00085 and 8.5σ above the global-fit value λ = 0.22501 ± 0.00068. The ratio does agree with tan θ_C = |V_us/V_ud|: with |V_ud| = 0.97367 ± 0.00032, the ratio is 0.2304 ± 0.0009 from the averaged |V_us| (3/13 lies 0.4σ above it) and 0.2311 ± 0.0004 from the K_μ2/π_μ2 determination of |V_us| (0.8σ below), and the global fit gives 0.2309 ± 0.0007 (0.2σ below). Both integers arise from the combinatorial topology of the Császár triangulation, but the construction derives a number, not the observable it should be compared with, and the tan θ_C reading was found only after the comparison with data. It is therefore recorded as a post-hoc observation (Section 9, Conjecture 3), not as a derivation or a prediction.

Note (unresolved conjecture). The formula |V_cb| = 3/[13(φ + 4)] ≈ 0.04108 reproduces the PDG 2022 value 0.0408 ± 0.0014 to 0.7%, where the suppression factor satisfies the algebraic identity $6 - 1/\varphi^2 = \varphi + 4$ (with 6 the vertex degree of K_7). This numerical agreement is documented for completeness. However, the golden ratio φ does not appear in the spectral data of K_7, the face adjacency graph eigenvalues, or the modular parameter of the Császár torus (ρ = e^{iπ/3}, involving √3 not √5). The suppression factor therefore lacks a prior address in the Császár geometry and the formula is not claimed as a derivation. See Open Problem (iii).

6.2 The Complex Structure Asymmetry

The mechanism producing the CKM hierarchy is the asymmetric action of the complex structure J = L_{e_0} on the up-type and down-type subspaces. Restricted to the up-type (outgoing) vertices {e_1, e_2, e_3}, J couples Generation 1 and Generation 3 ($J(e_1) = e_3$, $J(e_3) = -e_1$). Restricted to the down-type (incoming) vertices {e_6, e_5, e_4}, J couples Generation 2 and Generation 3 ($J(e_4) = e_5$, $J(e_5) = -e_4$). The up-sector rotation lives in the

(1,3) plane; the down-sector rotation lives in the (2,3) plane. Their product produces mixing in all three planes with the natural hierarchy |V_us| > |V_cb| > |V_ub|.

6.3 Open CKM Elements

The remaining matrix elements — |V_ub|, |V_td|, |V_ts|, and the CP-violating phase δ — have not been derived. The perturbation mechanism is identified: octonion non-associativity between the three quaternionic subalgebras creates inter-generation transition amplitudes. The associators are nonzero for half of the inter-generation triples (equal to $\pm 2e_0$) and zero for the other half. This asymmetry should encode the Jarlskog invariant, but the computation is incomplete.

7. The Rank-4 Resolution: Gauge Independence from Topological Dimension

The gauge group SU(3) × SU(2) × U(1) has rank 4. The automorphism group $G_2 = \mathrm{Aut}(\mathbb{O})$ has rank 2. This gap was identified by Krasnov (private communication) as the central obstruction to the present construction. We address it by showing that SU(2) and U(1) arise not from subgroups of G_2, but from the geometric and conformal structure of the Császár embedding, acting on topological data of different dimension than the vertex labels on which G_2 acts. Sourcing the additional rank outside $\mathrm{Aut}(\mathbb{O})$ is itself standard in the division-algebra program — algebraically from the $\mathbb{H}$ and $\mathbb{C}$ factors of $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ (Furey 2018a), or by triality enlargement of $\mathrm{der}(\mathbb{O})$ to $\mathrm{tri}(\mathbb{O}) = \mathfrak{so}(8)$ (Furey & Hughes 2025) — and the claim of this section is confined to the topological realization of that move on embedding data (see Section 11).

7.1 Three Gauge Actions on Three Topological Strata

The Császár polyhedron carries three geometrically independent types of data:

Stratum 0 (vertices). The seven vertices carry the imaginary octonion units e_1 through e_7. The automorphism group G_2 acts on these labels. Fixing one vertex reduces G_2 to its SU(3) stabilizer, acting on the six remaining vertices as $3 \oplus \bar{3}$. This is standard — the coset space $G_2/SU(3) = S^6$ has dimension 6, matching the vertex degree of K_7.

Stratum 1 (edges). The 21 edges of the Császár polyhedron are embedded as line segments in $\mathbb{R}^3$. Each edge carries a tangent vector and a normal plane. The orientation of the normal plane — the choice of normal and binormal directions — is the edge framing. Rotations of this frame form SO(3), and the spin structure induced by the $\mathbb{R}^3$ embedding lifts this to SU(2). The embedding in $\mathbb{R}^3$ has a unique spin structure — it is inherited from the ambient space, not selected from alternatives.

Identification with the weak SU(2) (conjecture). The existence of an SU(2) acting on edge framings establishes the gauge group factor but does not by itself identify it with the weak isospin SU(2) of the Standard Model. That identification requires two additional steps, both of which are conjectured here rather than derived. First, the coupling must be chiral: the SU(2) edge-framing action must couple to left-handed fermion modes only. The Császár embedding is intrinsically chiral (Section 8), and the conjecture is that this chirality selects the L-enantiomer as the one admitting a non-trivial SU(2) bundle extension, while the R-enantiomer encounters a topological obstruction. If this obstruction argument can be made precise — computing the relevant Stiefel-Whitney or framing class for each enantiomer — it would supply the V−A structure geometrically. Second, the SU(2) doublet structure must be connected to the generation structure of Section 4: each of the three quaternionic subalgebras $\mathbb{H}_1$, $\mathbb{H}_2$, $\mathbb{H}_3$ carries its own unit-quaternion group $\mathrm{Sp}(1) \cong \mathrm{SU}(2)$ (the double cover of $\mathrm{Aut}(\mathbb{H}) \cong \mathrm{SO}(3)$), and the weak SU(2) is identified with the diagonal or generation-averaged action

on the edge framings. The precise quotient map has not been constructed. Both steps are recorded as open problems (see Open Problem (xi)).

Stratum 2 (surface). The 14 triangular faces form a toroidal surface of genus 1. As a Riemann surface, the torus carries a complex structure with U(1) phase rotations. This conformal data is determined by the surface geometry and is independent of the vertex labeling and edge framing.

7.2 Independence and Commutativity

The three gauge transformations are geometrically independent:

Relabeling a vertex (SU(3)) permutes the octonion assignments without moving any edge in space or altering the surface complex structure. Rotating an edge frame (SU(2)) reorients the normal plane of that edge without changing which octonion labels sit at its endpoints or altering the global conformal structure. Rotating the surface phase (U(1)) changes the complex structure without affecting vertex labels or edge framings.

If the fermion representation space factorizes as a genuine tensor product V_color ⊗ V_weak ⊗ V_hypercharge — with SU(3) acting on the first factor, SU(2) on the second, and U(1) on the third — then gauge transformations acting on distinct factors commute automatically, and the total rank is rank(SU(3)) + rank(SU(2)) + rank(U(1)) = 2 + 1 + 1 = 4. This tensor factorization is not, however, automatic from the geometric independence of the strata. The vertices, edges, and faces of the Császár complex are not independent objects but the cells of a chain complex linked by the boundary maps $\partial_2 : C_2 \to C_1$ and $\partial_1 : C_1 \to C_0$ — an edge is a pair of vertices, a face a triple of edges — and cells of different degree are direct summands of this complex, not tensor factors. Under the color SU(3) the 21 edges accordingly carry a non-trivial representation: explicit computation on the octonion derivation algebra gives the su(3)-module decomposition $\Lambda^2(\mathbf{7}) = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{3} \oplus \bar{\mathbf{3}}$, in which the adjoint **8** appears and exactly one of the 21 dimensions is color-inert. The edge data is therefore not a color-singlet factor. The same conclusion follows independently from the algebraic side: for a normed division algebra W, $\mathrm{der}(W) \subset \mathrm{tri}(W)$ acts identically on the spinor, conjugate-spinor, and vector representations (Furey & Hughes 2025), so color, sitting inside $\mathrm{der}(\mathbb{O})$, acts alike on every triality representation — predicting from that direction too that the edge stratum cannot be color-inert. Whether the three geometric actions can nonetheless be realized on a common space as a genuine tensor product — on which they then commute identically — is a substantive requirement, not a structural triviality; it is recorded as Open Problem (i).

7.3 Why the Additional Rank Does Not Violate the G_2 Constraint

The rank-2 limitation of G_2 applies to transformations acting on the same representation space — the octonion vertex labels. The SU(2) edge framing and U(1) surface phase do not act on vertex labels. They act on geometric data that exists only because the Császár polyhedron is embedded in $\mathbb{R}^3$ as a Riemann surface, not merely defined as an abstract combinatorial object. The abstract K_7 graph has G_2 symmetry on its vertices and nothing else. The Császár embedding enriches K_7 with edge framings and a conformal structure that carry the additional rank.

This necessity can be made precise. Within $G_2 = \mathrm{Aut}(\mathbb{O})$ itself there is no room for a commuting weak factor: the centralizer of the color su(3) in g_2 is trivial. As an su(3)-module the Lie algebra g_2 decomposes as $g_2 = \mathbf{8} \oplus \mathbf{3} \oplus \bar{\mathbf{3}}$ (the classical branching; Günaydin & Gürsey 1973) — the adjoint of su(3) together with the $\mathbf{3} \oplus \bar{\mathbf{3}}$ tangent space of the coset $G_2/SU(3) = S^6$ — and this decomposition contains no copy of the trivial representation, so no non-zero element of g_2 commutes with all of su(3). The statement is verified by direct computation: solving the derivation condition for $g_2 = \mathrm{Der}(\mathbb{O})$, restricting to the 8-dimensional stabilizer of a fixed imaginary unit, and computing its centralizer in g_2 returns dimension zero. Consequently a second gauge factor commuting with color cannot be found inside G_2 under any choice of embedding, and the additional rank must be carried

by structure external to Aut(𝕆) — precisely what the edge framings and surface conformal structure supply. The rank-4 obstruction identified by Krasnov is in this sense intrinsic to G_2, not an artifact of the construction, and the stratification of §7.1 is the minimal response to it.

The dimension count $\dim(G_2) = \dim(SU(3) \times SU(2) \times U(1)) + \dim(H_1(T^2))$ gives $14 = 12 + 2$, where the two extra dimensions of G_2 beyond the Standard Model gauge group match the two generators of $H_1(T^2)$ — the topological data that supports the edge framings. This is a structural observation, not a derived identity: no map has been constructed sending the two extra G_2 generators to the two torus 1-cycles. The coincidence is noted as a consistency check on the stratification, and is recorded as an open derivation target (see Open Problem (x)).

7.4 The Spin Structure

An abstract genus-1 surface admits $2^{2g} = 4$ spin structures. However, a surface embedded in $\mathbb{R}^3$ does not freely choose among them. The ambient space $\mathbb{R}^3$ has a unique spin structure, and the embedding pulls it back to the surface canonically. The Császár polyhedron, as a physical object in three-dimensional space, inherits a specific spin structure from $\mathbb{R}^3$. That structure is even (Arf invariant 0): it is antiperiodic around two of the three nonzero classes of $H_1(T^2; \mathbb{Z}_2)$ and periodic around one. The all-periodic structure is the odd one (Arf invariant 1), and no embedding induces it. (Versions through v6.3 stated the reverse, that the alternatives to the induced structure are those with antiperiodic conditions around non-contractible cycles.)

Consequently, the SU(2) gauge action on edge framings is uniquely determined by the embedding. This is stronger than a selection among alternatives: the embedding forces a single spin structure with no free choice.

7.5 The Tensor Product Decomposition and the Dixon Algebra

The three topological strata provide a geometric realization of the Dixon algebra $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ (Dixon 1994) — the factor-to-force assignment $\mathbb{O} \to SU(3)$, $\mathbb{H} \to SU(2)$, $\mathbb{C} \to U(1)$ that runs through the modern division-algebra corpus (nearest statement: one Standard Model generation from a single copy of $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, Furey & Hughes 2022a; see also Furey & Hughes 2022b):

Vertices (0-dimensional) → 𝕆 → SU(3). The 7 vertices carry the imaginary octonion units. Fixing one vertex gives SU(3) as the stabilizer in G_2.

Edge framings (1-dimensional) → ℍ → SU(2). The 21 edges, viewed as framed curves in $\mathbb{R}^3$, carry quaternionic structure via the following explicit map. Each edge e carries an adapted orthonormal frame: the unit tangent T, a chosen unit normal N, and the binormal $B = T \times N$, together with a scalar density (the edge length or a local metric coefficient). These four quantities (T, N, B, scalar) are assigned to the quaternion basis as (i, j, k, 1) respectively. Under this identification the frame-rotation group SO(3) acting on (T, N, B) is the image of SU(2) under the standard 2:1 covering $SU(2) \to SO(3)$, with the scalar component carrying the U(1) norm. The action of SU(2) on the edge framing is therefore the spinor double cover of the SO(3) frame rotation, not merely an analogy. This identification is well-defined on each edge independently; commutativity with the vertex and surface actions holds once the tensor-product structure of §7.2 is established, which — as noted there — is itself an open requirement rather than an automatic consequence of the dimensional separation of the strata.

Surface complex structure (2-dimensional) → ℂ → U(1). The Császár torus as a Riemann surface carries a complex structure with U(1) phase rotations.

Granting that tensor-product structure, the gauge actions commute and the total rank is $2 + 1 + 1 = 4$. The additional rank beyond G_2 arises from the geometric and conformal structure of the Császár embedding, not

from subgroups of G_2; and, as established in §7.3, no weak factor commuting with color exists inside G_2, so an external source for the additional rank is mandatory rather than merely convenient.

8. Conjecture: Geometric Origin of Parity Violation

The Császár polyhedron has trivial symmetry group (C_1): it possesses no planes of symmetry, no rotational axes, and no center of symmetry. The absence of improper symmetries does not depend on the coordinates chosen: all 42 automorphisms of the triangulated torus (the group AGL(1,7) of Section 3) preserve the orientation of the surface, while a mirror plane, a center of symmetry or a rotoreflection axis would each induce an orientation-reversing automorphism, so no realization in $\mathbb{R}^3$ has any of them. It is therefore intrinsically chiral, existing as two non-superimposable enantiomers $\mathscr{M}$_L and $\mathscr{M}$_R.

Conjecture 1 (Chiral Coupling). *The SU(2) weak interaction, acting on the quaternionic edge-framing factor, is permitted only for one enantiomer of the Császár embedding. The L-enantiomer admits a vanishing obstruction class for the relevant bundle transition, while the R-enantiomer encounters a non-trivial topological obstruction enforcing its status as an SU(2) singlet.*

If correct, this provides a geometric origin for the V−A structure of the weak interaction. The relevant obstruction theory — involving ribbon framing, total torsion, and writhe of the K_7 edge graph — has not been carried out.

Adjacent prior art, differentiated. Parity violation has been obtained from the algebraic side of the division-algebra program: Furey (2018c) demonstrates that electroweak theory can violate parity automatically in the leptonic case, from ideal and conjugation structure, and Furey & Hughes (2022b) obtain left–right symmetric Higgs representations from quaternionic triality within a symmetry-breaking cascade. Conjecture 1 targets the same phenomenon by a distinct mechanism — a topological obstruction distinguishing the two enantiomers of a chiral embedding — and is neither derived from nor reducible to those algebraic mechanisms; no priority is claimed in either direction, and the obstruction computation remains open as stated.

9. Structural Predictions and Falsifiability

Verified claims:

(1) SU(3) arises as Stab_{G_2}(e_i), acting on the octonionic vertex structure.

(2) Three fermion generations correspond to the three Fano lines through any vertex. A fourth generation would falsify this.

(3) *Withdrawn in v6.4.* Earlier versions listed the hypercharges (0, +1/3, −2/3, +1) from Y = (−1)^(N+1) × N/3; that operator is withdrawn (Section 5). The charge spectrum of the three-mode Fock space is Furey's Q = N/3 (Furey 2015), and this paper's contribution there, the identification of the three modes with the three Hamiltonian 7-cycles of K_7, is a construction rather than a verifiable claim.

(4) The rank-4 gauge group arises from three independent topological strata of the Császár embedding, not from subgroups of G_2. That no weak factor commuting with color can instead be found inside G_2 is a theorem: the centralizer of the color su(3) in g_2 is trivial, since $g_2 = \mathbf{8} \oplus \mathbf{3} \oplus \bar{\mathbf{3}}$ as an su(3)-module contains no trivial summand (verified by direct computation on Der($\mathbb{O}$)). The commutativity of the three actions, by contrast, is contingent on establishing a genuine tensor-product realization of the representation space (Open Problem (i)) and is not claimed here as established.

(5) *Withdrawn from this list in v6.4.* Earlier versions listed the Cabibbo angle sin(θ_C) = 3/13 = 0.2308, within 2.6% of the PDG value. Against PDG 2024 that statement is excluded at 7.6σ–8.5σ (Section 6.1). The agreement of 3/13 with tan θ_C is listed below as Conjecture 3.

Conjectures (untested):

(1) The intrinsic chirality of the Császár polyhedron provides a geometric origin for parity violation.

(2) |V_cb| = 3/[13(φ+4)] = 0.04108, within 0.7% of PDG 2022. The golden ratio suppression factor lacks a prior address in the Császár geometry.

(3) *Post-hoc observation.* 3/13 = tan θ_C = |V_us/V_ud|, within 0.8σ of every PDG 2024 determination (Section 6.1). Both integers are combinatorial invariants of the Császár triangulation, but the construction does not select tan θ_C over sin θ_C, and this reading was identified after comparison with data. It becomes a result only if a derivation fixes the observable before the comparison.

10. Open Problems

The following are explicitly unsolved and are not claimed as results:

(i) Establishment of a genuine tensor-product realization V_color ⊗ V_weak ⊗ V_hypercharge of the three gauge actions. The commutativity asserted in Section 7.2 is automatic once such a factorization is in hand — operators on distinct tensor factors commute identically, so no separate bracket computation is required — but the factorization itself is not automatic: the strata are the cells of a chain complex rather than independent factors, and the 21 edges carry a non-trivial color representation (Section 7.2). The required content splits according to how the weak SU(2) is realized. (a) If SU(2) acts on the edge-framing fibre (the four quantities (T, N, B, scalar) per edge) while color acts on the edge labels, the two commute precisely when the framed-edge bundle is trivial — i.e. when the color action lifts to the bundle without frame holonomy. Because the bundle is associated to a non-trivial color representation, this is not automatic, and deciding it is exactly Open Problem (ii). (b) If SU(2) is the weak isospin acting on the octonionic fermion doublets, commutativity with color is the Furey–Dixon centralizer statement, which holds in $C\ell(6)$ because there SU(2) lives on the $\mathbb{H}$ factor of $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, genuinely separate from the $\mathbb{O}$ carrying color; the Császár-specific burden is then to show the edge-framing data faithfully reconstructs that $\mathbb{H}$ factor, which is Open Problem (xi)(b). In either reading the problem reduces to one already on this list, and is not an independent computation.

(ii) Computation of the obstruction invariant distinguishing the two Császár enantiomers under quaternionic framing extension.

(iii) Prior address for the golden ratio φ in the V_cb suppression factor. The spectral data of K_7 and its face adjacency graph do not contain φ. The modular parameter of the Császár torus is ρ = e^{iπ/3}, involving √3 not √5. The prior address, if it exists, must come from the algebraic structure (octonion multiplication, associators) rather than the combinatorial geometry.

(iv) The remaining CKM elements: |V_ub|, |V_td|, |V_ts|, and the CP-violating phase δ.

(v) The PMNS neutrino mixing matrix.

(vi) Mass hierarchies and coupling constant values from the lattice invariants.

(vii) Quantization of the classical vertex structure — connecting the combinatorial lattice to quantum field theory.

(viii) Whether the K_8 embedding on genus 2 carries F_4 or E_6 structure relevant to grand unification.

(ix) Which observable, if any, the ratio 3/13 determines. Read as sin θ_C it is excluded (Section 6.1); read as tan θ_C it agrees, but that reading is post hoc (Conjecture 3). Renormalization-group running cannot rescue the sin θ_C reading: in the Standard Model the Wolfenstein parameter λ changes only by $O(10^{-4})$ between the weak and Planck scales (Grossman et al. 2022), far below the 2.6% gap. (Versions through v6.3 asked instead whether such running could account for the overshoot.)

(x) Construction of an explicit map from the two extra generators of G_2 (beyond SU(3)) to the two generators of $H_1(T^2)$. The dimension count 14 = 12 + 2 is a structural consistency observation; it becomes a theorem only when this map is made explicit and shown to be equivariant under the full gauge action.

(xi) The left-handed coupling of the SU(2) edge-framing action. Two steps are required: (a) computation of the topological obstruction class distinguishing the L- and R-enantiomers of the Császár embedding under quaternionic framing extension, to establish that the SU(2) bundle couples exclusively to the L-enantiomer; and (b) construction of the quotient map identifying the weak isospin SU(2) with the diagonal action of $Sp(1)_1 \times Sp(1)_2 \times Sp(1)_3$ on the generation structure (the unit-quaternion groups of $\mathbb{H}_1$, $\mathbb{H}_2$, $\mathbb{H}_3$, each the double cover of $Aut(\mathbb{H}_i) \cong SO(3)$), rather than with each generation's SU(2) separately. Until both steps are completed, the identification of the edge-framing SU(2) with the Standard Model weak isospin SU(2) is a conjecture.

11. Relationship to Prior Work

The algebraic core of this construction is closely related to work by Furey (2014; 2015; 2016; 2018a,b; 2025), who showed that division algebraic ladder operators reproduce Standard Model fermion representations with correct gauge charges. Dixon (1994) introduced $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ as the algebraic design of physics; the present work proposes the Császár polyhedron as its geometric realization. Gürsey & Tze (1996) studied octonions in particle physics extensively. Krasnov (2018, 2024) approached the Standard Model gauge group from Spin(10) geometry. Boyle & Farnsworth (2020) explored non-associative geometry and the Pati-Salam model via Jordan algebras. The Császár polyhedron is well-studied in combinatorial topology; its connection to octonion multiplication via the Fano plane has not previously, to our knowledge, been used to derive gauge group structure or to propose values for quark mixing parameters.

Three-generation mechanisms, differentiated. Several distinct threefold mechanisms exist in the division-algebra program, and none coincides with the Fano point-valence mechanism of Section 4. Furey & Hughes (2025) identify the Standard Model symmetries inside tri(ℂ) ⊕ tri(ℍ) ⊕ tri(𝕆) and realize three generations as the triality triple (Ψ_+, Ψ_-, V) — spinor, conjugate-spinor, and vector representations of $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ — with the third generation extracted from the vector copy via a Cartan factorization; the three there is S_3 triality, not incidence valence. Furey (2014; 2018b) realizes three generations as three generation "prints" within the $C\ell(6)$ decomposition, with unbroken symmetry SU(3) × U(1) only. Gillard & Gresnigt (2019) obtain three generations from the complex sedenions, splitting $\mathbb{C} \otimes \mathbb{S}$ into three $\mathbb{C} \otimes \mathbb{O}$ subalgebras that share a common quaternionic subalgebra — structurally the nearest relative of the present mechanism, and still distinct: three octonion subalgebras through a common ℍ there, versus three quaternionic subalgebras $\mathbb{H}_1, \mathbb{H}_2, \mathbb{H}_3 \subset \mathbb{O}$ through a common unit here. The sedenion construction has since been extended to $C\ell(8)$, with the three generations carrying an explicit S_3 family symmetry (Gourlay & Gresnigt 2024) — again an automorphism three, not incidence valence. One further near-miss deserves explicit note: the decomposition $e_7 \cong (\text{tri}(\mathbb{H}) \oplus \text{tri}(\mathbb{O})) + 3(\mathbb{H} \otimes \mathbb{O})$ also contains three copies of $\mathbb{H} \otimes \mathbb{O}$, but those are triality-related copies of the *same* algebra — different objects, different threes. On the rank-4 problem, the principal algebraic alternatives to the topological-strata sourcing of Section 7 are rank from the ℍ and ℂ factors (Furey 2018a) and triality enlargement of der(𝕆) (rank 2) to tri(𝕆) = so(8) (rank 4), with the weak su(2)_L realized inside tri(ℍ) (Furey & Hughes 2025); as noted in Section 7, the external-sourcing move is standard in this program, and the present contribution is confined to its topological realization on embedding data. By contrast, no derivation of quark mixing angles

appears in this corpus — Yukawa couplings there remain free parameters — so the 3/13 ratio of Section 6.1 and the Fano$^+$/Fano$^-$ doublet-mismatch mechanism of Section 6 remain, to our knowledge, without counterpart.

12. Conclusion

The Császár polyhedron is selected by uniqueness: it is the only neighborly triangulation of the only $\chi = 0$ compact orientable surface. Its combinatorial structure produces SU(3) via vertex stabilization, three generations via Fano line valence, and a geometric carrier, the three Hamiltonian cycles, for the modes of Furey's number-operator charge. The rank-4 problem is addressed by the observation that the three gauge groups act on topological data of different dimension — vertex labels, edge framings, and surface complex structure — inheriting additional rank from the embedding geometry rather than from G_2 subgroups.

The ratio of the Fano valence to the rank of the boundary operator in the Császár triangulation, 3/13, built from two combinatorial invariants of the polyhedron, is excluded as sin θ_C by PDG 2024 and agrees with tan θ_C; because that reading was identified after the comparison with data, it is recorded as a post-hoc observation (Conjecture 3). The |V_cb| element is reproduced to 0.7% but with an algebraically unmotivated golden-ratio factor; it is documented as a conjecture pending a prior address.

Three claims are presented as verifiable results, two others having been withdrawn in v6.4 (Section 9); three conjectures, one of them a post-hoc observation, and eleven open problems are explicitly identified as unsolved. The identification of the edge-framing SU(2) with the weak isospin SU(2) is explicitly labeled as a conjecture pending the obstruction computation of Open Problem (xi). The construction is offered for independent mathematical verification.

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